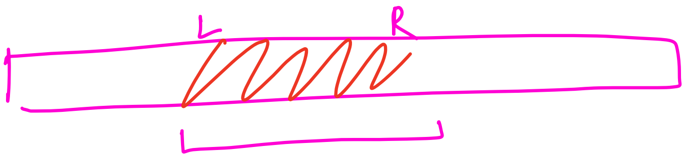
5 zeros
2 ones → 5 ones
2 zeros.

1 0 0 1 1 0 0 0 1 1 1 0 1



x 0s
y 1s → y 0s
x 1s ⇒ nr. of 1s changes from y to x

⇓

Result → Original nr. of 1s $+ (x - y)$.

13 0s
10 1s → $(13 - 10)$ 1's will be added.

Find the range where

nr. of 0s − nr. of 1s is maximum.

$0 \rightarrow 1$
$1 \rightarrow -1$ } Kadanes.

Q1) Given a row-wise and column-wise sorted matrix, find whether element k is present or not.

$$A = \begin{bmatrix} -5 & -2 & 1 & 13 \\ -4 & 0 & 3 & 14 \\ -3 & 2 & 6 & 18 \end{bmatrix}$$

$k = 13 \Rightarrow$ True

$k = 2 \Rightarrow$ True

$k = 15 \Rightarrow$ False

Brute force $\longrightarrow O(n * m)$ T.C

search all rows, all cols.

Binary search in every row $\longrightarrow O(n * \log m)$

OR

in every col $\longrightarrow O(m * \log n)$.

✓ decrease

dec

> 2

$$A = \begin{bmatrix} -5 & -2 & 1 & 13 \\ -4 & 0 & 3 & 14 \\ -3 & 2 & 6 & 18 \end{bmatrix}$$

increase

inc

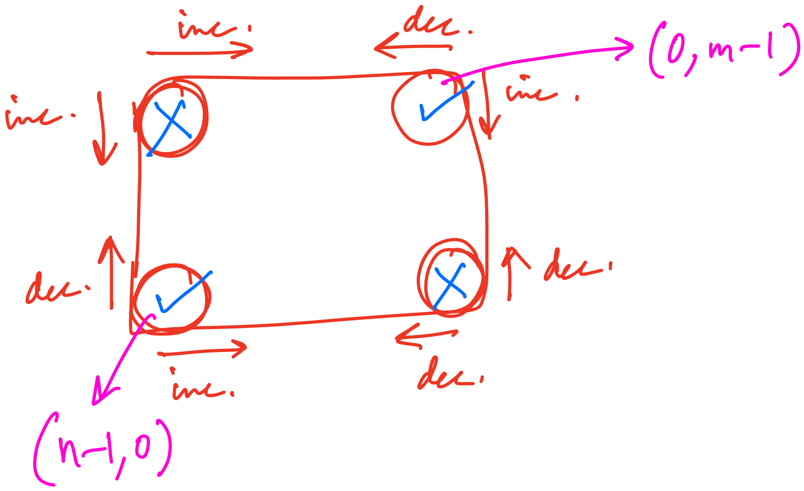
$\boxed{k = 2}$

$1 < 2$

```
i = 0, j = m-1
while (i < n && j >= 0) {
    if (arr[i][j] == k)
        return True
    if (k < arr[i][j])
        j = j-1        // Left
    else
        i = i+1        // Bottom
}
return False
```

$O(n+m)$ T.C,

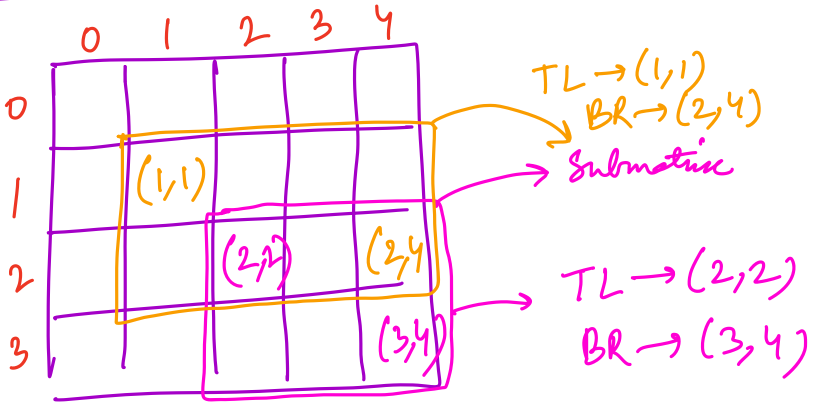
$O(1)$ S.C

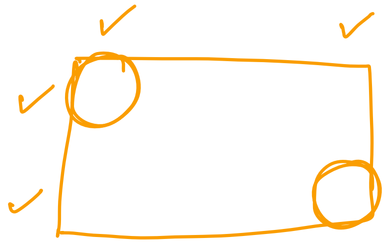


$$
\begin{array}{c|cccc}
 & j & & & \\
i & 0 & 1 & 2 & 3 \\
\hline
0 & \cancel{1} & \cancel{3} & \cancel{5} & 7 \\
1 & \cancel{8} & \cancel{8} & \cancel{10} & 12 \\
2 & \cancel{13} & \cancel{14} & \cancel{15} & 18 \\
3 & 19 & 22 & 25 & 30
\end{array}
$$

$k = 22$

Submatrix





TL, BR $\Longleftrightarrow$ r_1 r_2 c_1 c_2

- r_1 — starting row
- r_2 — ending row
- c_1 — starting col
- c_2 — ending col.

Q.2) Given a matrix A[n][m], find the sum of the sums of all submatrices of A.

$$\begin{bmatrix} 4 & 9 & 6 \\ 5 & -1 & 2 \end{bmatrix}$$

$[5] \rightarrow 5$

$[5 \ -1] \rightarrow 4$

$[5 \ -1 \ 2] \rightarrow 6$

$[-1] \rightarrow -1$

$[-1 \ 2] \rightarrow 1$

$[2] \rightarrow 2$

$\begin{bmatrix} 9 \\ -1 \end{bmatrix} \rightarrow 8$

$\begin{bmatrix} 9 & 6 \\ -1 & 2 \end{bmatrix} \rightarrow 16$

$\begin{bmatrix} 6 \\ 2 \end{bmatrix} \rightarrow 8$

$[4] \rightarrow 4$

$[4 \ 9] \rightarrow 13$

$[4 \ 9 \ 6] \rightarrow 19$

$[9 \ 6] \rightarrow 15$

$[9] \rightarrow 9$

$[6] \rightarrow 6$

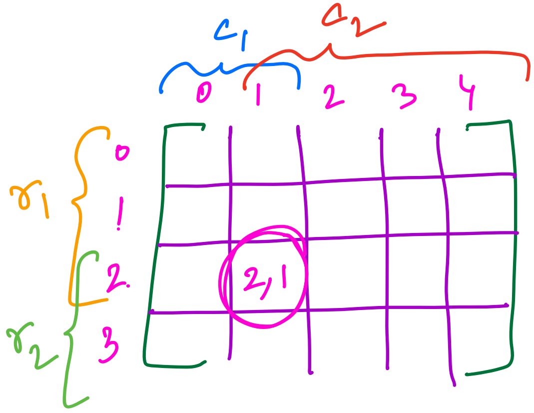
$\begin{bmatrix} 4 \\ 5 \end{bmatrix} \rightarrow 9$

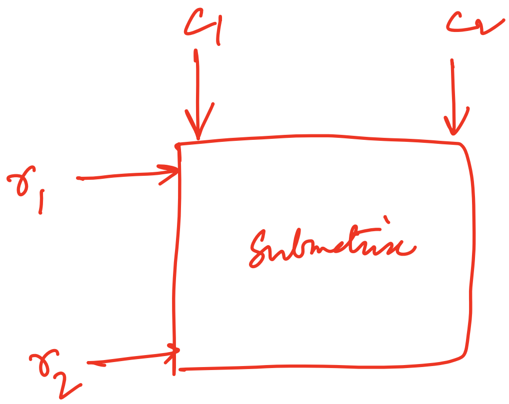
$\begin{bmatrix} 4 & 9 \\ 5 & -1 \end{bmatrix} \rightarrow 17$

$\begin{bmatrix} 4 & 9 & 6 \\ 5 & -1 & 2 \end{bmatrix} \rightarrow 25$

$-1 * 8$

+ ;





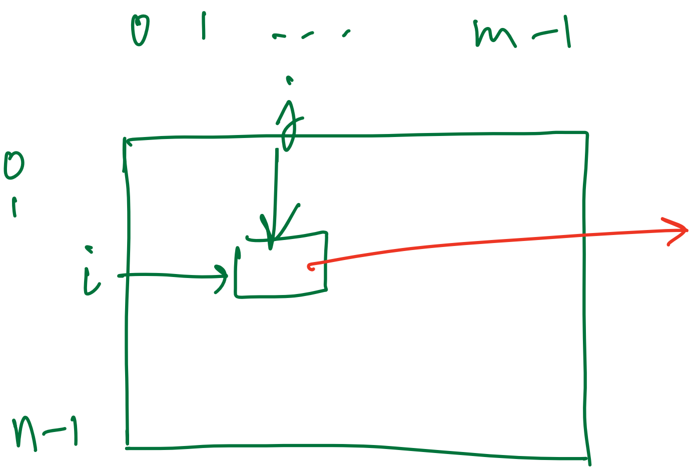
$r_1 \to [0,2] \to 3$

$r_2 \to [2,3] \to 2$

$c_1 \to [0,1] \to 2$

$c_2 \to [1,4] \to 4$

$3*2*2*4 = 48.$

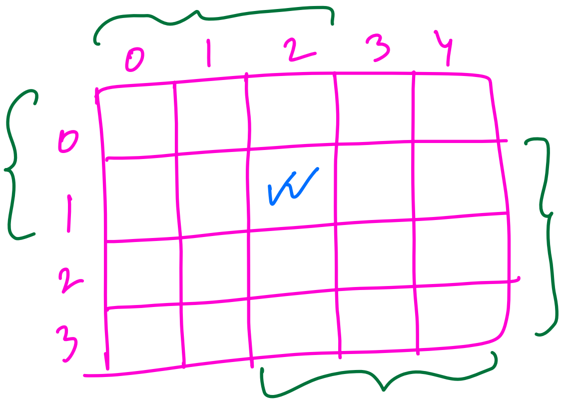


how many submatrices?

		#
$r_1 \to$	$[0, i]$ $\to$	$i+1$
$r_2 \to$	$[i, n-1]$ $\to$	$n-i$
$c_1 \to$	$[0, j]$ $\to$	$j+1$
$c_2 \to$	$[j, m-1]$ $\to$	$m-j$

No. of submatrices that contain $(i,j) \longrightarrow (i+1)*(n-i)*(j+1)*(m-j)$

$$\text{Sum} = \sum_{\substack{i \in [0, n-1] \\ j \in [0, m-1]}} \left[arr[i][j] * (i+1) * (n-i) * (j+1) * (m-j) \right]$$



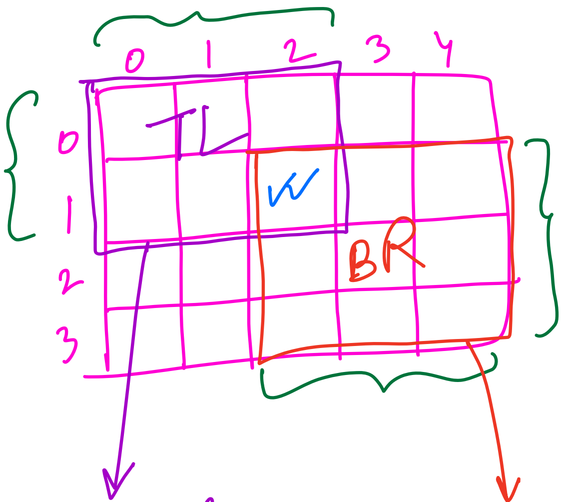
$i=1, j=2$

$n=4$

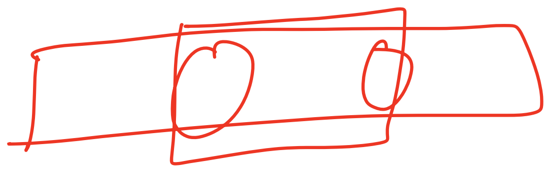
$m=5$

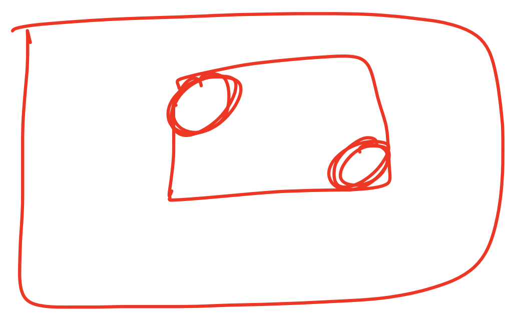
$$\underset{\downarrow\,2}{(i+1)} * \underset{\downarrow\,3}{(n-i)} * \underset{\downarrow\,3}{(j+1)} * \underset{\downarrow\,3}{(m-j)}$$

54.



$(i+1)*(j+1)$ $(n-i)*(m-j)$





No. of submatrices = No. of TL cells * No. of BR cells

$= (i+1)*(n-i)*(j+1)*(m-j)$

[Break till 10:40 PM]

Q3) Given arr[n], find the first missing natural no. in arr.

$\{3, -2, 1, 2, 7\} \longrightarrow 4.$

$\{-9, 2, 6, 4, -8, 1, 3\} \longrightarrow 5$

$\{-2, 6, 2, 5, 9, 4, 3, 8\} \longrightarrow 1$

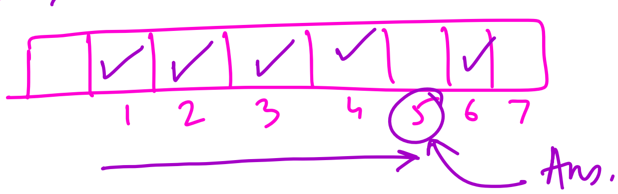
Brute Force $\longrightarrow$ Try from $1, 2, 3, \dots n, n+1$.

$O(n^2)$ T.C.

Sort $\longrightarrow$ $O(n \cdot \log n)$ T.C.

Array of size $n+1$ to store Present/Absent $\longrightarrow$ $O(n)$ T.C., $O(n)$ S.C

$\{-9, 2, 6, 4, -8, 1, 3\}$



Dict./Hash Map $\longrightarrow$ $O(n)$ T.C., $O(n)$ S.C

$\{-3, -1, -2, \boxed{6}, -5\}$

1 2 3 4 5

4 (Ans)

$\{-3, -1, -2, 6, -5\}$

0 1 2 3 4

idx

$3+1=4$.

1	→ 0
2	→ 1
3	→ 2
4	→ 3
5	→ 4

$\{3, -2, 1, 2, 0\}$ $n=5$

$n+1=6$

0 1 2 3 4

$\{-3, -6, -1, 2, 6\}$

$3+1=4$.

Ans.

6 6

0 1 2 3 4 5 6 7 8 9

$\{2, -1, 3, 4, 9, 3, 1, -6, 6, -5\}$

0 1 2 3 4 5 6 7 8 9

$\{-2, -11, -3, -4, 9, -3, 1, 11, -6, 11\}$

$4+1=5$.

$-|-3| = -3$

```
for (i -> 0 to n-1) {
    if (A[i] <= 0)
        A[i] = n+1
}
for (i -> 0 to n-1) {
    x = abs(A[i])
    if (x-1 <= n-1) {
        A[x-1] = -1 * abs(A[x-1])
    }
}
ans = n+1
for (i -> 0 to n-1) {
    if (A[i] > 0) {
        ans = i+1
        break
    }
}
return ans
```

$\boxed{\begin{array}{l} O(n)\ T.C \\ O(1)\ S.C \end{array}}$